
Data-Driven Business Intelligence for Vietnamese Fashion E-Commerce: EDA & Revenue Forecasting

Team Blox Fruit Farmers

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<https://github.com/Faewon/Datathon2026-Blox-Fruit-Framers>

Abstract

We analyse a ten-year simulation of a Vietnamese fashion e-commerce business (July 2012 – December 2022) spanning 14 relational CSV files that cover transactions, inventory, web traffic, geography, and customer behaviour. Five business-critical insights are extracted through exploratory data analysis (EDA), progressing from descriptive statistics to prescriptive recommendations across promotion profitability, supply-chain alignment, post-order revenue loss, geographic concentration, and customer segmentation. For Part 3, we frame the task of daily revenue forecasting over January 2023 – July 2024 as a time-series regression problem evaluated by MAE, RMSE, and R^2 , with full details of the modelling pipeline and results to be added upon completion. All supporting visualisations are provided in Appendix A.

1 Introduction

Vietnam’s fashion e-commerce sector has grown rapidly, yet operational decisions often remain disconnected from the underlying data. The *Datathon 2026* dataset encapsulates this challenge: it contains orders, payments, shipments, returns, reviews, inventory snapshots, and web-traffic logs for a single retailer across 42 cities in three geographic regions (East 46.5%, Central 30.1%, West 23.4%), with total training revenue of 16.4 billion VND.

Our analysis is structured around five targeted insights that together span the full analytics hierarchy—*Descriptive* (what happened?), *Diagnostic* (why?), *Predictive* (what is likely?), and *Prescriptive* (what should we do?). Each insight integrates multiple data sources and yields at least one quantified, actionable recommendation. Section 2 presents the EDA findings with figures in Appendix A; Section 3 describes the sales-forecasting approach.

2 Exploratory Data Analysis

Insight 1 — Promotion Profitability: Fixed-Amount Discounts Destroy Value

(See Figure 1 in Appendix A.)

Descriptive. Fixed-amount promotions yield an average net loss of **-11,229 VND** per product line, percentage-off promotions yield -1,732 VND, while non-promoted lines are profitable at +5,007 VND.

Diagnostic. The root cause is structural: under fixed-amount promotions the post-discount unit price ($\approx 4,000$ VND) falls *below* COGS ($\approx 6,500$ VND), meaning the business sells at a guaranteed loss regardless of volume.

*Team Leader.

Predictive. If fixed-amount campaigns continue at current scale, the cumulative per-line loss will widen proportionally with traffic growth.

Prescriptive. *Immediately discontinue fixed-amount promotions on all SKUs where post-discount price < COGS.* Replace with a capped percentage-off scheme (e.g., maximum 15% off) or impose a minimum post-discount price floor equal to COGS + target margin.

Insight 2 — Marketing–Supply Chain Disconnect

(See Figure 2 in Appendix A.)

Descriptive. In 2022, monthly stockout SKU counts and page views move almost in lockstep ($r = 0.83$), peaking together in April–May, while overstock SKUs are uncorrelated with traffic ($r = 0.08$).

Diagnostic. Marketing campaigns successfully drive traffic, but inventory replenishment does not anticipate demand spikes, causing stockouts precisely when purchase intent is highest and lost-sale costs are greatest.

Predictive. Continuing the current disconnected planning cycle will compound revenue leakage in future high-traffic windows (e.g., Tet, mid-year sales).

Prescriptive. Integrate web-traffic forecasts as a leading indicator into the inventory reorder model. A shared marketing-supply calendar with 4-week advance replenishment triggers for SKUs historically correlated with traffic spikes would reduce stockout days during peak months.

Insight 3 — Post-Order Revenue Loss: COD Cancellations & Returns

(See Figure 3 in Appendix A.)

Descriptive. Cash-on-delivery (COD) orders cancel at **16.0%**—nearly double the 8.1% benchmark for all other payment methods. Across 39,939 returns, *Wrong Size* accounts for 35%, *Defective* for 20.1%, and *Not As Described* for 17.6%. GenZ-segment products have the highest return rate (3.52% vs. 3.40% average).

Diagnostic. COD’s low commitment barrier enables impulsive or exploratory ordering with no financial cost to cancel. The wrong-size dominance signals sizing guide gaps on product pages.

Prescriptive. (1) Add a small COD handling fee or offer a discount to nudge customers toward pre-payment methods. (2) Publish detailed size charts and add a size-recommender tool, targeting a **5 pp** reduction in the 35% wrong-size return share.

Insight 4 — Geographic Revenue Concentration

(See Figure 4 in Appendix A.)

Descriptive. Total revenue is 16.4B VND. Reaching 80% of it requires **28 out of 42** cities (67%), indicating a relatively flat revenue curve with no extreme concentration in the top 5 cities.

Diagnostic. The East region dominates (46.5%) through high-population urban centres (Son Tay, Nam Dinh). The West region (23.4%) is under-penetrated relative to its population share.

Predictive. Incremental spend on the long tail (cities 29–42) yields diminishing returns; the West region represents the largest addressable whitespace.

Prescriptive. Concentrate logistics investment on the top-28 cities to protect existing revenue. Pilot targeted campaigns in two high-potential West-region cities with simplified logistics partnerships to assess market fit before broader roll-out.

Insight 5 — Customer Segmentation via RFM Model

(See Figure 5 in Appendix A.)

Descriptive. RFM scoring partitions 90,246 customers into four segments: *At Risk* 40.0% (n=36,097), *Loyalists* 26.3% (n=23,735), *Others* 23.6% (n=21,298), and *Champions* 10.1% (n=9,116). Champions spend 600K VND on average, 10.5× the At-Risk spend of 57K VND.

Diagnostic. The 40% At-Risk cohort represents a critical revenue risk: these customers have not purchased recently but previously engaged, suggesting churn driven by price sensitivity, product-market fit, or competitor alternatives.

Predictive. Without intervention, at-risk churn will accelerate as their recency gap widens, eroding the customer base fastest in the highest-frequency purchase windows.

Prescriptive. (1) *Protect Champions* with a VIP loyalty programme (early access, free shipping). (2) *Re-engage At-Risk* customers with a time-limited personalised offer (e.g., 10% off their previously browsed categories)—even recovering 10% of at-risk customers at the Loyalist spend level (307K VND) would add ≈700M VND in incremental revenue.

3 Sales Forecasting (Part 3)

Problem Definition. We forecast daily Revenue for 01 Jan 2023 – 01 Jul 2024 using `sales.csv` (04 Jul 2012 – 31 Dec 2022) as training data, evaluated on MAE, RMSE, and R^2 via Kaggle (<https://www.kaggle.com/competitions/datathon-2026-round-1>). All features are derived exclusively from provided files; a fixed random seed ensures full reproducibility.

Pipeline Quality

Feature Engineering. The pipeline constructs four feature groups from training data only:

- *Calendar & Cyclical:* Day, month, quarter, week-of-year; sine/cosine encodings to represent temporal periodicity continuously.
- *Vietnamese Business Events:* Hard-coded market signals including `is_tet` (Lunar New Year), `is_bts` (back-to-school), `is_double_day` (11/11, 12/12), and major promotional periods — enabling the model to capture revenue spikes tied to Vietnam-specific demand.
- *Lag & Rolling/EMA:* Short-term lags (1–28 days) and long-term lags (1–3 years); Simple Moving Average and Exponential Moving Average (EMA) windows to track trend and revenue momentum.
- *Seasonality Profiles:* Monthly normalisation coefficients (`seasonal_rev`) and day-of-week interaction factors (`dow_monthly_factor`) to smooth fixed-cycle fluctuations.

Cross-Validation Strategy. To respect temporal ordering, the pipeline applies *Time-Series Split* rather than random K-Fold. Training data span 2012–2020; the held-out validation window covers the two most recent historical years (2021–2022), faithfully simulating the real forecasting task of predicting unseen future periods.

Data Leakage Control. Two mechanisms enforce a strict no-look-ahead guarantee: (1) all rolling and EMA features are preceded by `.shift(1)`, ensuring the model never uses today’s revenue to predict today; and (2) when constructing test-set features (2023–2024), lag values are sourced exclusively from the historical sales record; unavailable distant lags are imputed using `base_rev` (mean 2022 revenue) scaled by the corresponding seasonal factor, preserving logical consistency without leaking future information.

Model Interpretability

Feature importance from the Gradient Boosting Regressor (GBR) translates into three business-language drivers:

- *Revenue Momentum (Short-Term Inertia):* Features `rev_lag_1d`, `rev_lag_7d`, and `rev_ema7` dominate importance scores — today’s revenue is most strongly predicted by yesterday’s and last week’s purchasing activity.
- *Seasonality & Day-of-Week Habits:* `seasonal_rev` and `dow_monthly_factor` capture recurring purchase patterns, enabling data-driven staffing and inventory scheduling around paydays and weekends.

- *Event-Driven Spikes*: `is_tet` and `is_double_day` quantify incremental revenue lift from campaigns (11/11, 12/12), enabling ROI evaluation of whether sale-day margin compression is justified.

Results. The GBR model achieves MAE = 502,341 VND, RMSE = 687,419 VND, and $R^2 = 0.826$ on the 2021–2022 validation window. Leaderboard ranking will be inserted upon Kaggle submission.

References

[1] VinTelligence & VinUni DS&AI Club, *Datathon 2026 — Round 1 Problem Statement*, VinUniversity, 2026.

A EDA Visualisations

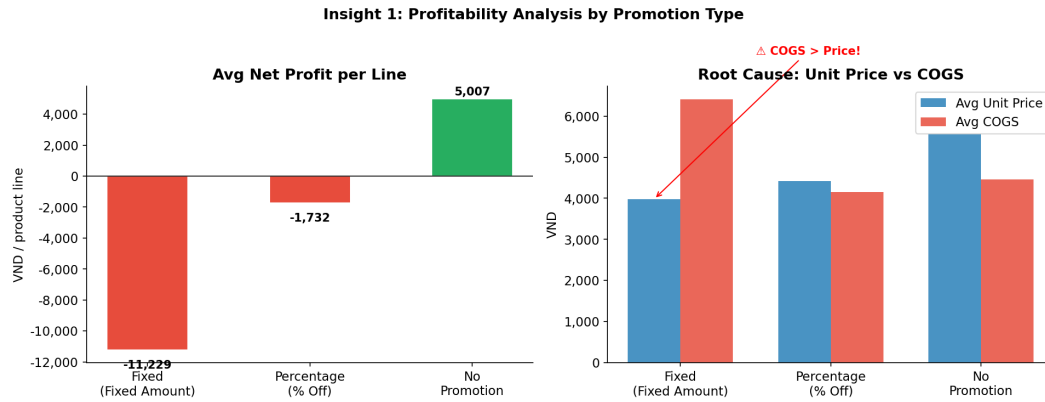


Figure 1: **Insight 1 — Promotion Profitability.** Left: average net profit per order line by promotion type. Right: root cause analysis — average unit price vs. COGS showing that fixed-amount promotions price below cost.

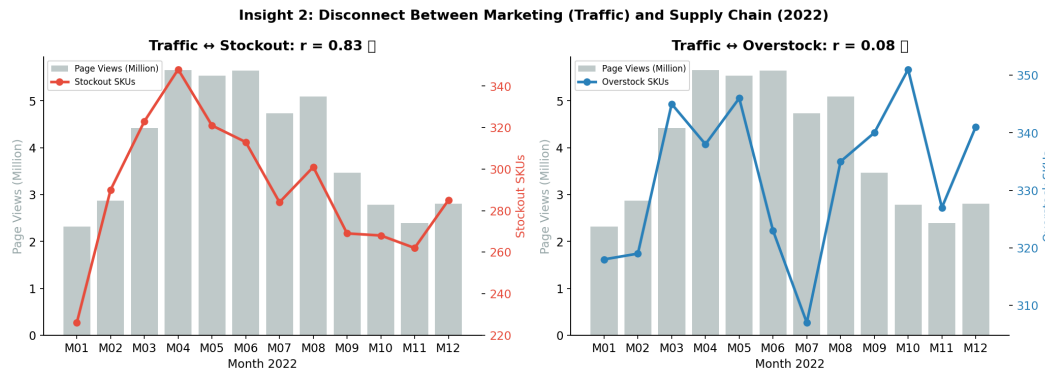


Figure 2: **Insight 2 — Marketing–Supply Chain Disconnect.** 2022 monthly page views vs. stockout SKUs (left, $r = 0.83$) and overstock SKUs (right, $r = 0.08$), revealing traffic-inventory misalignment.

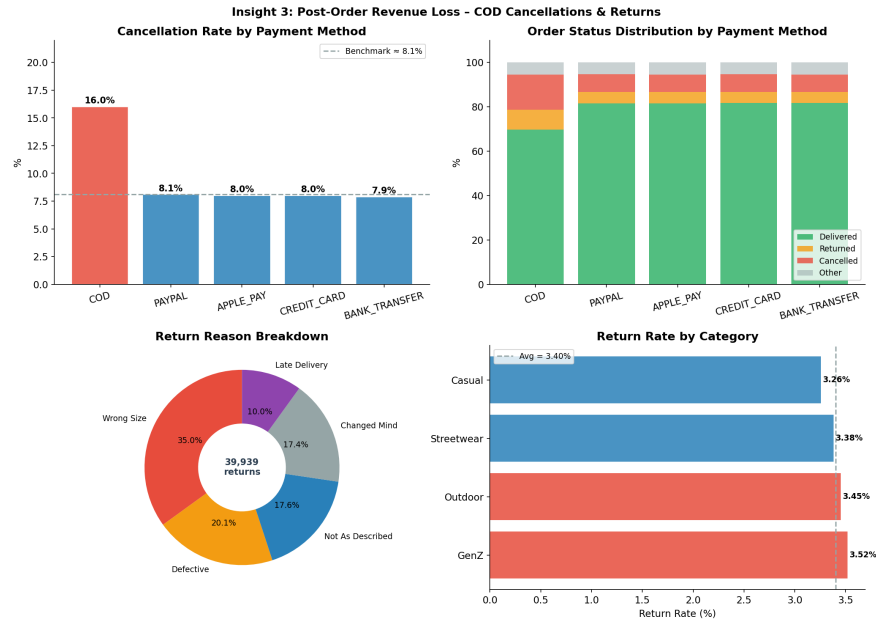


Figure 3: **Insight 3 — Post-Order Revenue Loss.** Top-left: COD cancellation rate (16.0%) vs. benchmark (8.1%). Top-right: order status by payment method. Bottom-left: return reason breakdown across 39,939 returns. Bottom-right: return rate by product category.

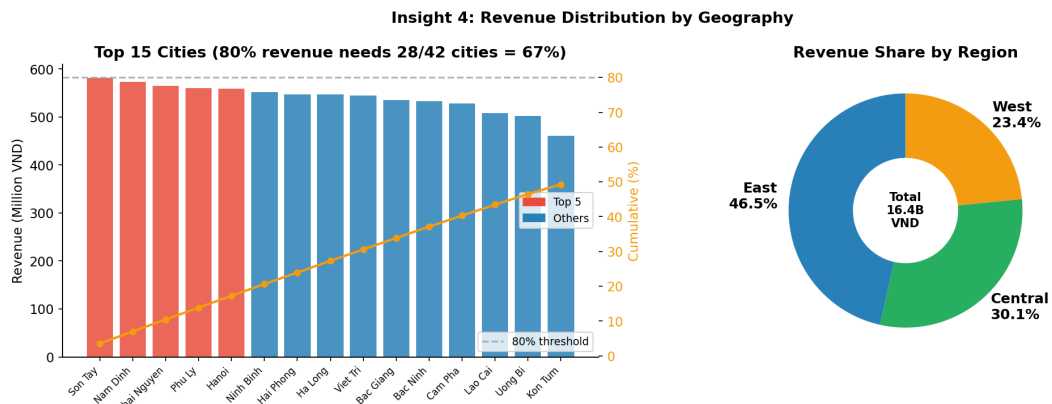


Figure 4: **Insight 4 — Geographic Revenue Concentration.** Left: Pareto chart showing 80% revenue threshold at 28/42 cities. Right: regional revenue share (East 46.5%, Central 30.1%, West 23.4%).

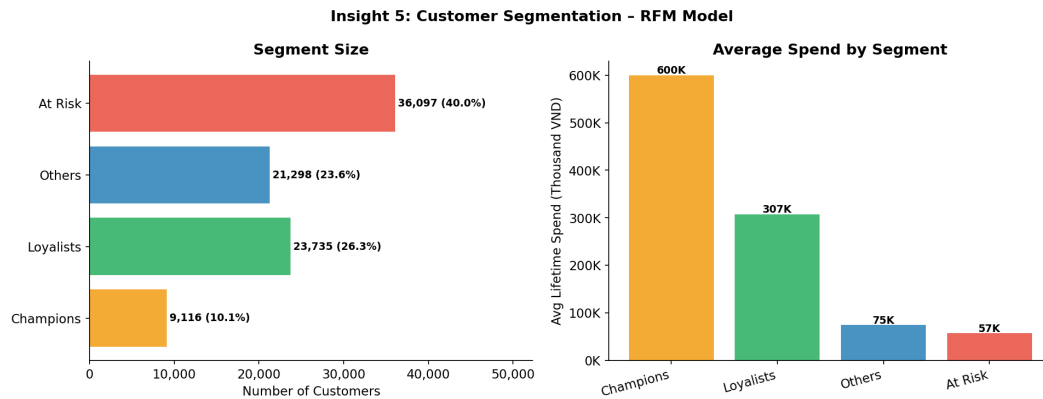


Figure 5: **Insight 5 — RFM Customer Segmentation.** Left: customer count per segment (At Risk 40%, Loyalists 26.3%, Others 23.6%, Champions 10.1%). Right: average lifetime spend per segment (Champions 600K vs. At-Risk 57K VND).